

ODROBNĚJ: POZITIVNÍ DEFINITNOST

$$\begin{aligned}
 \frac{1}{4} (\|m+n\|^2 - \|m-n\|^2) &= \frac{1}{4} (\langle m+n, m+n \rangle - \langle m-n, m-n \rangle) = \frac{1}{4} (\langle m, m \rangle + \langle m, n \rangle + \langle n, m \rangle + \langle n, n \rangle - \langle m, m \rangle + \langle m, n \rangle - \langle n, m \rangle - \langle n, n \rangle) \\
 &= \frac{1}{2} (\langle m, n \rangle + \langle n, m \rangle) = \frac{1}{2} (\langle m, n \rangle + \overline{\langle m, n \rangle}) = \operatorname{Re}(\langle m, n \rangle) \\
 \frac{1}{4} (\|m-in\|^2 - \|m+in\|^2) &= \frac{1}{4} (\langle m-in, m-in \rangle - \langle m+in, m+in \rangle) = \frac{1}{4} (\langle m, m \rangle + \langle m, -in \rangle + \langle -in, m \rangle + \langle -in, -in \rangle - \langle m, m \rangle - \langle m, in \rangle - \langle in, m \rangle - \langle in, in \rangle) \\
 &= \frac{1}{4} (-i \langle m, n \rangle + \overline{-i \cdot \langle m, n \rangle} - i \langle m, n \rangle + \overline{-i \cdot \langle m, n \rangle}) = \frac{1}{2} (-i \langle m, n \rangle + i \overline{\langle m, n \rangle}) = \frac{1}{2} (-i \operatorname{Re}(\langle m, n \rangle) - i \operatorname{Im}(\langle m, n \rangle) + i \operatorname{Re}(\langle m, n \rangle) + i \operatorname{Im}(\langle m, n \rangle)) \\
 &= \frac{1}{2} (-i \operatorname{Im}(\langle m, n \rangle) - i \operatorname{Im}(\langle m, n \rangle)) = -i \operatorname{Im}(\langle m, n \rangle) \cdot i = \operatorname{Im}(\langle m, n \rangle) \\
 &\text{podle } \operatorname{Im}(\bar{a}) = -\operatorname{Im}(a)
 \end{aligned}$$